



ME 519: Technical communication for Engineers

Course Code	ME 519
Course Name	Technical communication for Engineers
L-T-P-C	0-2-0-2
Prerequisites	None
Intended for	MTech / MS / PhD in Engineering
Mutual Exclusion	None
Approval	30th Senate (Feb 2021), in the M.Tech Fluid and Thermal Engineering (FTE) curriculum; 44th BoA (Nov 2021)

Course Contents

- **Introduction:** What is technical writing and how it is different from non-technical writing. Standard writing models, building blocks and their description, and order of writing, case studies: establishing a clear distinction between “good” and “bad” writing (group activities are recommended), avoiding flowery language, word-processing tools (Latex). Writing assessment checklist. [4Lectures]
- **Writing introduction section:** Structure of the introduction section, literature review and referencing, grammar (active/passive, signalling language), sentence structures and paragraphing, model development, testing model on relevant papers/reports, vocabulary (establishing significance, verbs for previous work, identifying gap, the present work). [5Lectures]
- **Methodology:** Introduction, grammar and writing skills (tense pairs, ‘a’ vs ‘the’, countable and uncountable nouns, adverbs), model building, model testing exercise, vocabulary (general overview of the methods section, essential background information, providing precise details about materials and methods, justifying the choices made, taking appropriate care, comparison to other studies). [2Lectures]
- **Results:** Importance of reporting objectively, correct use of tenses, the importance of sequence, comparison with previous work, model development, model testing, vocabulary (existing research, general overview, invitation to view results, key results, sequence, frequency, quantity, causality, comparison with other results, problems with results, implications). [3Lectures]
- **Discussion and conclusion:** Structure, grammar and sentence structure (ability, possibility, probability, certainty, opinion, obligation), modals, model development, model testing, vocabulary (revisiting previous sections, summarising key results, refining implications, relationship to existing research, achievement/contribution, limitations, future work, applications). [4Lectures]
- **Abstract and title:** Abstract models, grammar and writing skills (choice of verb tense), length, language, model development, model testing, vocabulary (background, aim, problem, paper description, methods, results, achievements, implication, limitation), title and keywords. [2Lectures]
- **Technical presentations:** Data analysis and interpretation, specification of uncertainty, choice of scales, data density, difference between graphs for a manuscript and graph for a ppt, structure, tools, graphics, practice presentations. [8Lectures]

Text Books

1. Glasman-Deal, H., **Science Research Writing for Non-native Speakers of English**, World Scientific, 2010.

References

1. McCaskill, M. K., **Grammar, Punctuation and Capitalization**, NASA, Scientific and Technical Information Division, Washington DC, 1990.
2. Vidoli, C. A., **Technical Report Writing**, NASA Technical Memorandum 105419.
3. Paradis, J. G. and Zimmerman M.L., **The MIT Guide to Science and Engineering Communication**, Cambridge, Mass.: MIT Press, 1997.